Uncertainty Signal Pilot: Perception and Reasoning Entropy

FERMAT Handwritten Math, Qwen2.5-VL-3B/7B, LLaVA-NeXT, and InternVL3

August 11, 2026

Abstract

This study tests whether two entropy-based uncertainty signals – *perception entropy* (instability in transcribing a handwritten math answer) and *reasoning entropy* (instability in grading whether an answer contains an error) – predict model correctness, on the FERMAT dataset. It ran in seven phases: an exploratory $n = 100$ pilot on Qwen2.5-VL-3B, a confirmatory $n = 300$ run with a pre-registered analysis and a deliberately rebalanced label distribution, a follow-up closing the two gaps that run left open, a fourth phase testing Qwen2.5-VL-7B on the reasoning arm specifically, a fifth phase testing a second model family (LLaVA-NeXT), and a sixth phase testing a third model family (InternVL3). Perception is unambiguous on Qwen; reasoning is more specific than either a clean positive or a clean negative. **The Phase 4–5 stratified reasoning result was subsequently withdrawn as an evaluation artifact — see the Retraction below, which supersedes every reasoning claim in this abstract and in §7–§8.** Phase 5 also finds a real limit on how far the perception result can currently be tested. Phase 6 adds a cautionary methodological finding: a third model's perception AUROC can look like the best result in the study and not survive the same robustness check every other perception number here is held to.

Perception entropy works. At $n = 300$ it reaches AUROC = 0.835, 95% CI $[0.787, 0.879]$, replicating the pilot's 0.788 and tightening the interval by half. It survives every artifact check: 0.796 $[0.736, 0.852]$ after removing all parse failures *and* all maximum-entropy items simultaneously. A simple decision rule falls out of it – when all five transcriptions disagree, the model is wrong 57 times out of 61 (93.4% precision). It also beats an independent baseline: the model's own token log-probabilities reach only AUROC 0.537, a resolved paired gap of +0.297 $[+0.214, +0.380]$ (§6.1), so the signal is not simply recoverable from information the model already exposes. A further five samples per item (§6.1) push $K = 10$ to AUROC 0.806 against $K = 5$'s 0.761 on a fair, like-for-like label, a resolved +0.045 $[+0.005, +0.087]$.

Reasoning entropy fails, for a reason that is not about entropy. The pilot's apparent 0.618 was an artifact of FERMAT's 87/13 class imbalance: a constant "there is an error" predictor scores 0.87 there, and the model's 75% grading accuracy was losing to it. Rebalancing to 50/50 at $n = 300$ removes that flattery and the model's accuracy collapses to **51.7% against a 0.500 baseline**. It answers "there is an error" on 93% of items – 94.7% correct on error-containing answers, 8.7% on clean ones. Reasoning entropy correspondingly measures nothing: AUROC = 0.522 $[0.462, 0.582]$, with 42% of items pinned at zero entropy. The finding is not that the metric is badly implemented but that *sampling entropy cannot predict correctness for a model performing at chance* – there is no competence for disagreement to track. A pre-registered stratified hypothesis, formed post-hoc on the pilot data, proved untestable: the model's bias left only 8 of 150 error items misgraded, below the registered minimum. Three alternative grading prompts were screened against this bias and none fixed it (§6.3); one of them swung the model's response rate from 94% "error" to 38% while leaving accuracy essentially unchanged, evidence that the ceiling is a capability limit rather than a promptable default.

A larger model does not rescue the reasoning arm, and the analysis that appeared to show it did was invalid. Qwen2.5-VL-7B's grading accuracy (59.0% vs. a 50.0% baseline)

clears a pre-registered capability gate only marginally, and its pooled AUROC (0.520) is statistically identical to 3B's. Stratifying by whether the graded answer actually contains an error appeared to produce confirmed AUROCs of 0.834 $[0.768, 0.891]$ at 7B and 0.854 $[0.796, 0.902]$ at 3B, replicated at 0.775 $[0.694, 0.848]$ on a second model family. **Those numbers are arithmetically correct and their interpretation was wrong.** Within a stratum where every item carries the same true label, "was the model correct" is the same column as "what did the model answer" — they agree on 100% of items at 7B — and the entropy is computed from the very votes that produce that answer, so the measurement is near-circular. A signal-free biased coin reaches a *higher* AUROC than any model here (null medians 0.901, 0.868, 0.831 against observed 0.854, 0.801, 0.775). The apparent sign reversal between strata is the same identity with its sign flipped: in the clean stratum correctness is the exact complement of the verdict, so the two stratum AUROCs sum to one. **The honest reasoning result is the pooled one on a balanced sample: ∼0.52, no signal.** Perception is unaffected, and § explains why — the difference is label cardinality, not label type.

The methodological lesson is separable from any single result. Adding bootstrap confidence intervals late overturned several conclusions the pilot had already formalized into code and prose; the single most consequential design decision was not the metric or the model but the *label balance of the evaluation set* – and Phase 4 shows the same lesson applies a second time, one level up: a real effect survived being pooled away only once the pooling itself was diagnosed and undone.

A second, architecturally different model family (Phase 5) exposes a real limit on the perception result — and reproduced the reasoning artifact rather than confirming a reasoning effect. LLaVA-NeXT (`llava-v1.6-mistral-7b-hf`) cannot be used to test perception entropy at all: its free-response transcription accuracy is 0.8% (2/243 items) against Qwen's ∼42%, a capability gate of the same kind Phase 4 applied to reasoning, now applied to perception for the first time. Its reasoning arm is unaffected by that gate and, once powered the same way as Phase 4, reached AUROC 0.775 $[0.694, 0.848]$ on 34 misgraded items. That was read at the time as the project's first cross-family replication; it is better understood as the same degeneracy reproducing on a third architecture, which is why replication did not catch it (Retraction).

A third model family (Phase 6, InternVL3-8B) produces this project's clearest case yet for why every perception result here is checked against a sensitivity cut. Its raw perception AUROC, 0.915 $[0.880, 0.944]$, is higher than Qwen's confirmed 0.835 despite InternVL3's base transcription accuracy (21.0%) being about half Qwen's (∼42%). Independently recomputing every derived column from the raw samples finds zero mismatches – this is real model behavior, not a scoring bug – but the same artifact cut every other perception number in this report survives (excluding maximum-entropy items) collapses InternVL3's AUROC to **0.556** $[0.495, 0.614]$**, indistinguishable from chance**. The mechanism: 67% of items (202/300) pin at exact maximum entropy, and within that group accuracy is a 0.99% floor, versus 62.2% among the rest – a bimodal coherent-vs-incoherent split, not a graded uncertainty signal. **The 0.915 headline must not be read as InternVL3 beating Qwen on perception.** Its reasoning arm is a smaller, first-of-its-kind result: the `has_error`=1 stratum (0.628 $[0.548, 0.706]$, 45 misgraded items) is adequately powered and its CI excludes chance, but does not clear this report's 0.70 confirmation threshold – a new, in-between verdict this report has not needed before.

Retraction (2026-08-09)

The `has_error`=1 stratified reasoning result reported in §7 and §8 — AUROC 0.775–0.854**, described there as confirmed across three model families — is withdrawn. It is an artifact.** Those sections have not yet been rewritten; read them only as a record of what was believed. The corrected treatment is in `paper/main.tex`.

Within a stratum in which every item carries the same ground-truth label, "was the model correct" is not a separate fact from "what did the model answer": the two are the same column, agreeing on 100% of items in the 7B run. Reasoning entropy is computed from precisely the votes that produce that answer, so the stratified AUROC is near-circular. A signal-free null — a model answering "error" with fixed probability per sample, carrying no item-level information at all — attains a *higher* AUROC than any model tested here: null medians 0.901, 0.868 and 0.831 against observed 0.854, 0.801 and 0.775.

The sign reversal between strata, presented in §7 as the central finding, is the same identity with its sign flipped: in the clean stratum correctness is the exact complement of the verdict, so the two stratum AUROCs sum to 1.

None of the safeguards this report relies on caught it. The pre-registered 0.70 threshold is cleared by the null's 2.5th percentile. Both strata were adequately powered. Replication across three model families reproduced the artifact three times rather than corroborating an effect.

What stands. The perception result (§5, AUROC 0.835) is unaffected, and the reason is structural: transcription correctness depends on *which* label wins rather than on how concentrated the votes are, so agreement does not imply correctness — 38 items are unanimous and 3 of those are wrong. The honest reasoning result is the pooled one on a balanced sample, ≈ 0.52, no signal. Phase 7's gate (§10) is reinforced: an all-erroneous dataset is a single-label stratum throughout.

Locked in `pilot/tests/test_stratum_degeneracy.py`.

1 Goal

Given a vision-language model transcribing and grading handwritten math answers, does the entropy of its own repeated outputs – how much it disagrees with itself across independent samples – predict when it is actually wrong?

The work ran in seven phases, and the distinction matters for how much weight any individual number carries:

- **Phase 1 ($n = 100$, exploratory, Qwen-3B).** Whether either signal exists at all. Sections 3–4. Several analyses here were chosen after seeing data, and are labelled as such.
- **Phase 2 ($n = 300$, confirmatory, Qwen-3B).** Section 5. A rebalanced sample with thresholds registered before the run (`pilot.plotting.SCALEUP_PREREGISTRATION`). This is the phase the headline perception claim rests on. Includes a risk-coverage and conformal deferral analysis of the $n = 300$ result (§5.6).
- **Phase 3 (same $n = 300$ items, follow-up, Qwen-3B).** Section 6. Closes two gaps Phase 2 left open: an independent baseline for perception entropy, and a screen of alternative grading prompts for the reasoning arm.
- **Phase 4 ($n = 300$ then $n = 650$, Qwen-7B).** Section 7. Tests whether a larger model changes the reasoning-arm conclusion: a pre-registered capability gate, a stratified re-analysis, and a targeted extension that resolves the stratum Phase 2 could not test, on both Qwen sizes.
- **Phase 5 ($n = 300$ then $n = 550$, LLaVA-NeXT).** Section 8. Tests whether a *different model family*, not just a different size, changes either conclusion: a capability gate for perception (it does not clear it) and a powered stratified re-analysis for reasoning (it confirms, and replicates Qwen-7B's magnitude).

- **Phase 7 ($n = 100$, ScratchMath, Qwen-7B).** Section 10. The only dataset change in this report. A sizing check, gated: the model answers "error" regardless of whether it can read the image, so the run measures response bias rather than grading.

- **Phase 6 ($n = 300$, InternVL3-8B).** Section 9. A third model family, trained independently of both prior families. Perception clears the accuracy bar this time (unlike LLaVA-NeXT) but its raw AUROC does not survive this report's standard sensitivity cut; reasoning's `has_error`=1 stratum is powered and resolved above chance but does not clear the confirmation threshold.

Still out of scope: no human double grading, no full-dataset run, no model larger than 7B/8B, and no second dataset/task. What Phase 4 removed from this list was whether reasoning entropy could work at all on a model with some real grading competence; what Phase 5 removed was whether either result is specific to the Qwen2.5-VL family (reasoning: no, confirmed; perception: untested, gated by a capability gap in the one alternative model tried so far); Phase 6 adds a model where perception could be measured but the headline number turned out not to be trustworthy without the sensitivity cut this report already applies everywhere else.

Every AUROC below is reported with a 95% bootstrap confidence interval (10,000 item-level resamples, seed 0; `pilot.plotting.bootstrap_auroc_ci`). These were added late, after the substantive experiments were complete, and they materially changed what this report can claim – several differences it originally presented as findings turn out to be within sampling error at $n = 100$. Where two conditions are compared, the comparison uses a *paired* bootstrap over the same resampled items (`bootstrap_auroc_difference_ci`) rather than checking whether two marginal intervals overlap, which is a strictly weaker and commonly misread test.

2 Setup

- **Data**: FERMAT (2244 items, 84.8% containing an error). Phase 1 drew 100 items at random (seed 42), inheriting that 87/13 split. Phase 2 drew 300 items *stratified* to 150/150 on `has_error` (`pilot.data.load_fermat_balanced`) – for reasons developed in §4.5, this turned out to be the single most consequential design choice in the study. Phase 4 reuses that same 300-item sample for its capability gate, then extends it with 200 more `has_error`=1 items only, disjoint by construction (`pilot.data.load_fermat_extra_error_items`), to power a stratum the 300-item sample could not measure.

- **Model**: Qwen2.5-VL-3B-Instruct (Phases 1–3); Qwen2.5-VL-7B-Instruct, full bf16, reasoning arm only (Phase 4, §7); LLaVA-NeXT (`llava-v1.6-mistral-7b-hf`), full bf16, both arms attempted but perception gated (Phase 5, §8); InternVL3-8B (`OpenGVLab/InternVL3-8B-hf`), full bf16, both arms attempted, perception measurable but not trustworthy at face value (Phase 6, §9).

- **Perception**: the model transcribes a handwritten Question–Answer pair; $K = 5$ samples at temperature 0.7, plus 2 samples at temperature 0 as a determinism sanity anchor.

- **Reasoning**: the model grades whether the handwritten Answer contains an error (binary 0/1); same $K = 5 + 2$ temperature-0 sampling scheme.

- **Entropy**: Shannon entropy in nats over the cluster distribution of the K sampled outputs for one item, $H = -\sum_i p_i \ln p_i$.

- **Correctness labels**: transcription is compared against the ground-truth answer; grading is compared against the ground-truth error label.

3 Phase 1: Perception Entropy – Two Failures, One Fix

3.1 Attempt 1: exact-string clustering (failed)

The transcription prompt asks for the *full* handwritten derivation, not just a final answer. Clustering by exact string match after light normalization treated any paraphrasing of that derivation as a different answer.

Items pinned at maximum entropy ($\ln 5 \approx 1.609$)	83/100

83% of items showed the maximum possible disagreement regardless of whether the model actually agreed with itself on the substance of the answer – the metric carried essentially no information.

3.2 Attempt 2: naive semantic clustering (failed)

Reasoning that exact-match was too strict, the next attempt clustered by semantic similarity instead: bidirectional NLI entailment and, separately, embedding cosine similarity, both applied to the whole derivation.

	Canonical (Attempt 1)	NLI	Embedding
Mean entropy	1.535	0.137	0.185
Items at max entropy	83/100	0/100	0/100

This looked like a fix – entropy dropped sharply – but the correctness rate that came with it (majority-vote transcription matched against ground truth via the same NLI entailment check) was 92/97 = 95%, implausibly high next to the model's other performance figures. Inspection showed the cause: with premise and hypothesis sharing nearly identical scaffolding (the same LaTeX structure, the same boilerplate phrasing), the NLI model's well-documented lexical-overlap bias pushed it toward "entailment" regardless of whether a buried number or coefficient actually differed. The fix for Attempt 1's problem introduced a new, opposite failure mode: over-merging.

3.3 Attempt 3: final-answer extraction (worked)

The actual defect in both prior attempts was comparing the *wrong span of text*: the full derivation, when only the final answer matters for a correctness judgment. Attempt 3 extracts just the final answer – the last multiple-choice option statement, a `\boxed{}` result, the last display-math block, or (rarest) the last line – before clustering with the same syntactic/canonicalization approach as Attempt 1.

Metric	Value
AUROC (perception entropy → transcription error)	**0.788**
95% CI	$[0.697, 0.869]$
Items at max entropy	14/100
Transcription correct	42/100
Median entropy, correct group ($n = 42$)	0.500
Median entropy, incorrect group ($n = 58$)	1.332
Mean entropy, correct group	0.583
Mean entropy, incorrect group	1.099

This is the strongest result in the pilot: real, visible separation between correct and incorrect items (Figure 2, left panel), and the only arm whose confidence interval excludes chance by a clear margin.

3.4 Sensitivity: is 0.788 an artifact?

An entropy AUROC can be manufactured two ways that have nothing to do with the model being usefully uncertain, and both are live concerns here. First, parse failures receive their own cluster label, so an unparseable sample inflates entropy *and* tends to be scored incorrect – the metric would then be measuring the parser rather than the model. Second, $K = 5$ entropy takes only 7 distinct values, and the 14 items pinned at $\ln 5$ are *all* incorrect; if that single cell carried the ranking, there would be no graded signal underneath it. Each cut was rerun with its own CI (`pilot.plotting.auroc_sensitivity`):

Subset	n	AUROC	95% CI
Full	100	0.788	$[0.697, 0.869]$
Excluding all parse failures	85	0.773	$[0.672, 0.865]$
Excluding max-entropy items	86	0.721	$[0.611, 0.822]$
Excluding both	74	**0.707**	$[0.587, 0.819]$

Neither cut explains the result. Dropping all 15 parse-failure items moves the estimate by 0.015, and only 3 of the 14 max-entropy items had a parse failure at all (mean entropy 0.99 for parse-failure items vs. 0.864 for clean ones – a small gap, not a confound). Under the harshest simultaneous cut the interval still excludes 0.5. The underlying distribution is monotone rather than concentrated in one cell: of items at zero entropy 11/14 were transcribed correctly, falling steadily to 0/14 at maximum entropy.

That top cell is worth stating as a result rather than treated only as a robustness threat: **every one of the 14 items where all five transcriptions disagreed was wrong** – precision 1.00 as an abstention rule, recall 0.24 of all 58 errors. On $n = 100$ a zero-false-positive cell of size 14 is suggestive rather than established, but it is the most directly actionable thing in the pilot.

4 Phase 1: Reasoning Entropy – Clean Pipeline, Unresolved Signal

Unlike perception, the reasoning/grading pipeline required no comparable fix. It clusters a single parsed binary digit (error present/absent) rather than free text, so it never had the "wrong span" problem. The pipeline was independently re-verified by recomputing correctness from scratch for all 100 items against the stored results: zero mismatches.

4.1 $K = 5$ baseline

Metric	Value
AUROC (reasoning entropy → grading error)	0.618
95% CI	$[0.489, 0.741]$
Grading accuracy	75%
Median entropy, correct group ($n = 75$)	0.5004
Median entropy, incorrect group ($n = 25$)	0.5004

This interval includes 0.5. With 25 items in the minority class, an AUROC of 0.618 is not distinguishable from chance, and earlier drafts of this report describing it as a "real but weak" signal overstated it. Everything downstream in this section was motivated by trying to strengthen a number that was never established in the first place.

The medians are also *exactly* tied. With only 5 samples of a binary outcome, there are just 3 realistic entropy values (unanimous, a 4–1 split, a 3–2 split), which structurally limits how well two groups can be told apart by this statistic.

4.2 $K = 15$ follow-up: is the weak signal a real ceiling, or under-sampling?

To test whether $K = 5$ was simply too coarse to reveal a real effect, grading was resampled to $K = 15$ for the same 100 items (reusing the original 5 samples and drawing 10 more), keeping perception untouched.

Metric	$K = 5$	$K = 15$
AUROC	0.618	0.665
95% CI	$[0.489, 0.741]$	$[0.514, 0.800]$
Grading accuracy	75%	82%
Distinct entropy values reached	5	13
Median entropy, correct group	0.500	0.439
Median entropy, incorrect group	0.500	0.608

The exact median tie *does* break at $K = 15$ (Figure 3), and the $K = 15$ interval does clear 0.5 – though only just. A pre-registered decision rule (AUROC ≥ 0.72 *and* the median tie breaking, to count as "sharper"; 0.55–0.72 as "real but modest"; below 0.55 as "noise") classifies this as **confirmed_modest**.

That classification should be read with the paired comparison in mind, because the apparent $0.618 \to 0.665$ gain is *not* evidence that more sampling helped: the paired difference is -0.047 with a 95% CI of $[-0.178, +0.090]$, comfortably spanning zero. The tie breaking is a genuine structural consequence of having 13 reachable entropy values instead of 5, but tripling K bought no measurable discrimination. Note also that grading accuracy itself moved (75% → 82%), so the two AUROCs are scored against different correctness labels – 25 errors vs. 18 – which the paired interval cannot correct for.

This pointed at a different question: is more sampling the right lever at all, or is the *digit itself* too coarse a readout of the model's judgment?

4.3 Reasoning-text clustering: a promising readout, an unproven improvement (unresolved)

The grading prompt asks the model to write a `**Reasoning:**` explanation before its `**Error:**` digit. Every analysis so far discarded that explanation and kept only the digit. Clustering the reasoning text instead – via the same bidirectional NLI entailment approach that failed for perception (Attempt 2) – was tested against the already-collected $K = 5$ raw samples, at no additional GPU cost.

Metric	Digit-only	Reasoning-text
AUROC	0.618	**0.702**
95% CI	$[0.489, 0.741]$	$[0.579, 0.817]$
Median entropy, correct group ($n = 75$)	0.500	0.950
Median entropy, incorrect group ($n = 25$)	0.500 (tied)	1.332
Paired difference (text − digit, same items, same labels): $+0.083$, 95% CI $[-0.064, +0.233]$		

The improvement is not statistically established. Reasoning-text clustering at $K = 5$ does clear chance on its own terms ($[0.579, 0.817]$ excludes 0.5) where the digit-only baseline does not ($[0.489, 0.741]$ does not) – but that pattern is precisely the "difference in significance is not a significant difference" trap. The correct test is the paired one, and its interval spans zero. An earlier draft of this report called this "a stronger, less noisy signal" and "arguably the most interesting result of the reasoning-arm investigation"; on the evidence available at $n = 100$, that claim is not supported. What survives is narrower and still worth having: reasoning-text clustering is the only reasoning-arm readout whose own interval excludes chance, and it has a documented mechanism (below) for why it should be less noisy than the digit. That makes it the right candidate to test at larger n, not a demonstrated win.

Given that the same NLI approach caused Attempt 2's over-merging failure on perception, this result was not trusted without directly checking for the same bias. The check: do any two samples that *disagree on the parsed digit* (one says error, one says no error) ever get merged into the same reasoning-text cluster? They do – 55 times across the 61 items where samples disagreed on the digit. Inspecting a diverse sample of these merges (8 cases, different items) showed 7 of 8 were not opposed reasoning merged by mistake: *both* texts explicitly argued "there is an error," just citing different specific details, while their own digits disagreed. Quantified across the full dataset: **60/494 samples (12%)** have digit$= 0$ while their own reasoning opens by flagging a mistake ("incorrect," "error," "mistake," ...). The model is self-contradicting within single samples – writing a reasoning that argues for an error, then emitting the opposite digit anyway, plausibly decoding noise on the final token, independent of the fully-formed judgment already written. At $K = 5$, reasoning-text clustering is not an artifact of NLI bias; it is a readout that is not vulnerable to this specific noise source the way the digit alone is (Figure 4).

This 12% self-contradiction rate is a measurement about the model, not about the clustering method, and it is unaffected by everything the confidence intervals overturn – it is a direct count over 494 parsed samples, not an AUROC comparison. It stands as an independent finding regardless of whether reasoning-text entropy ever proves to be the better predictor.

4.4 $K = 15$ GPU confirmation: the $K = 5$ result does not generalize (failed)

A GPU-accelerated confirmation at $K = 15$ (impractical on local CPU – a single item's 105 pairwise NLI comparisons took 35s locally; GPU reduced the full 100-item pass to 58s) was run via a dedicated notebook (`pilot/02_reasoning_text_entropy.ipynb`). It reproduced the $K = 5$ result almost exactly (AUROC 0.7019 vs. the 0.702 found locally), then revealed the $K = 15$ result does not hold up:

Metric	$K = 5$	$K = 15$
AUROC (reasoning-text)	0.702	**0.520**
95% CI	$[0.579, 0.817]$	$[0.378, 0.664]$
Items with a digit split	61/100	87/100
Cross-digit cluster merges	55	**1187**
Paired difference ($K = 5 - K = 15$, same items): $+0.182$, 95% CI $[+0.037, +0.330]$		

The $K = 15$ interval includes 0.5, so the technique is not distinguishable from chance at that K. More usefully, the degradation from $K = 5$ to $K = 15$ is the *only* reasoning-arm comparison in this report whose paired interval excludes zero – this is the one place where the data can actually tell two conditions apart. It carries one caveat that the interval cannot absorb: grading accuracy differs between the two runs (25 errors vs. 18), so the two AUROCs are scored against different label sets. The mechanistic evidence below is what makes the degradation credible rather than the interval alone.

The paired comparison against the digit-only baseline at $K = 15$ is -0.145, 95% CI $[-0.317, +0.037]$: directionally worse, not resolvably so. The cross-digit merge count grew far faster than the $\sim 10\times$ increase in available pairs ($\binom{15}{2} = 105$ vs. $\binom{5}{2} = 10$) would predict – a $\sim 21\times$ increase in merges from a $\sim 10\times$ increase in comparisons. Re-clustering individual items locally to inspect actual cluster membership found the cause: **transitive cascade merging**. The clustering method unions any two samples whose reasoning mutually entails, then merges transitively (union-find) – so a single mistaken pairwise link can silently absorb an entire subgroup into a cluster that does not actually agree. One item's largest cluster contained 11 of 15 samples, mixing digit=1 samples arguing *"incorrectly multiplies by 5 instead of 9/5"* with digit=0 samples arguing *"does not contain any apparent...errors"* – genuinely opposed conclusions, merged together. This is not universal: a second inspected item's largest cluster (7/15 samples) was entirely digit=0 with consistent reasoning, correctly merged. The failure is concentrated in a subset of items, but at $K = 15$ that subset is large enough to collapse the aggregate signal to near-random. More pairwise comparisons mean more chances for a single false-positive entailment call to cascade, and union-find has no mechanism to catch or bound that once it happens.

The mechanism is established: the cascade was found by re-clustering real items and reading actual cluster membership, and it reproduces as a unit test. What the confidence intervals add is a bound on how much of the *aggregate* $0.702 \to 0.520$ drop can be attributed to it – the paired interval $[+0.037, +0.330]$ is wide, so the degradation is real in direction but poorly determined in magnitude. A structural fix (e.g. requiring a minimum fraction of pairwise agreement within a cluster rather than any transitive chain, or a stricter/larger NLI model) would need to precede any further use of this technique at higher K – though at $n = 100$ such a fix could not be evaluated with any precision either, which is why sample size is the recommended next lever rather than the algorithm.

4.5 Manual inspection: why the reasoning arm underperforms

The confidence intervals establish that the reasoning signal is unresolved but say nothing about *why*. To separate the three plausible causes – clustering, prompt ambiguity, and the model simply being confidently wrong – individual failure cases were read directly, selected from the two informative quadrants: low entropy but graded wrong, and high entropy but graded right.

The grading task is close to degenerate. The 100-item sample is **87/13** on the ground-truth `has_error` label. A constant "there is an error" predictor therefore scores **0.87**, while the model's majority vote scores 0.75 at $K = 5$ and 0.82 at $K = 15$. *The model loses to the trivial baseline.* The model also answers "error" in 383 of 495 parsed samples and for 86 of 100 items, and identifies only **1 of the 13 clean answers** correctly. Grading accuracy is therefore not a meaningful measure of task competence here, and an uncertainty metric layered on top of it inherits the problem.

The dominant failure is confident wrongness, not clustering. Of the 7 items with low entropy (< 0.3) that were graded wrong, 6 have a clean ground-truth answer: the model invented an error and all five samples agreed. Entropy cannot flag these – there is no disagreement to measure. Reading the 6 unanimous cases, they are arithmetic and recall failures rather than defensible disagreements: on $2x^2 - 4x + 3 = 0$ the model asserts $b = -9$ and computes a discriminant from it; on the distance between $(1, 7)$ and $(4, 2)$ it states the formula should be $\sqrt{(1 + 4)^2 + (7 - 2)^2}$; in another it declares $2ab = 29-9$ incorrect and then derives the same value the answer gave. Only one of the six resembles prompt ambiguity (objecting to the *method* used to eliminate a variable rather than to the result) – and even there the model's proposed alternative does not work. **Clustering is not implicated in any of them**: the digit-only pipeline clusters an atomic 0/1 token, and these items are unanimous.

Pooling hides a signal that works. Because the model's bias runs one way, entropy's predictive direction reverses between strata (`pilot.plotting.stratified_auroc`):

Configuration	Stratum	n	n_{wrong}	AUROC (95% CI)
Digit, $K = 5$	pooled	100	25	0.618 $[0.489, 0.739]$
	has error (gt=1)	87	13	**0.756** $[0.618, 0.876]$
	clean (gt=0)	13	12	0.125 $[0.000, 0.250]$
Digit, $K = 15$	pooled	100	18	0.665 $[0.512, 0.799]$
	has error (gt=1)	87	6	**0.891** $[0.813, 0.955]$
	clean (gt=0)	13	12	0.000 $[0.000, 0.000]$
Text, $K = 5$	pooled	100	25	0.702 $[0.579, 0.817]$
	has error (gt=1)	87	13	0.770 $[0.635, 0.890]$

Within the majority stratum the digit-only signal is considerably stronger than anything reported elsewhere in this document, and it *improves* with K ($0.756 \to 0.891$) – reversing, within this stratum, the pooled finding that $K = 15$ bought nothing. Within the 13 clean items it is inverted: the model is confidently wrong there, and the single item it graded correctly is the one where it was most uncertain. Pooling two strata whose predictor runs in opposite directions does not produce a weaker version of the real effect; it produces a different quantity that describes neither.

This is a hypothesis, not a result. The stratification was chosen after seeing the pooled numbers, so it is post-hoc subgroup analysis and is not protected by the confidence intervals shown. The $K = 15$ majority stratum rests on only 6 misgraded items, where bootstrap intervals are optimistic, and the 13-item clean stratum is far too small to estimate anything from. The defensible claim is narrower than the table looks: *the reasoning metric's weakness is attributable to the model's systematic error-present bias on a severely imbalanced label, not to the entropy measure or its clustering*, and the stratified comparison should be pre-registered and re-run on the larger sample rather than reported as a finding now.

Consequence for the study design. A grading task where 87% of items share one label cannot support this question well regardless of n. The scale-up should rebalance the grading subset toward roughly 50/50 on `has_error` – otherwise a larger sample buys tighter intervals around a quantity that is still mostly measuring class imbalance.

5 Phase 2: The $n = 300$ Balanced Run

Phase 1 left two questions it could not answer. Four of its five comparisons were unresolvable at $n = 100$, and its most interesting reasoning-arm claim – that entropy works within the has-error stratum and is merely cancelled by pooling – was a subgroup analysis chosen after seeing the data, resting on 13 clean items. Phase 2 was designed to settle both.

5.1 Design

Two changes from Phase 1, plus a pre-registration:

- $n = 100 \to 300$, taking the 95% CI half-width on a single AUROC from about ± 0.12 to about ± 0.06.
- **87/13 → 50/50 on `has_error`.** A census of FERMAT (2244 items, 340 clean) confirmed a balanced sample of this size was feasible. Balancing does two things at once: it removes the confound of §4.5, and it *increases power*, because the model over-predicts "error" and so clean items are the ones it gets wrong. A 50/50 sample at $n = 290$ buys the same precision as an unbalanced $n = 480$.
- **Thresholds fixed before the run** (`SCALEUP_PREREGISTRATION`, registered 2026-08-02), including the minimum minority-class size of 30 below which no verdict is read at all. This is what converts Phase 1's post-hoc stratified claim into a real test.

5.2 Perception: replicated and tightened (worked)

	$n = 100$	$n = 300$	Verdict
Pooled	0.788 [0.697, 0.869]	**0.835** [0.787, 0.879]	
`has_error`=1 stratum	0.762 [0.662, 0.855]	**0.830** [0.760, 0.893]	**replicated**
Clean stratum	0.940 [0.762, 1.000] *(13 items)*	0.839 [0.772, 0.898]	

The like-for-like comparison is the `has_error`=1 stratum, since the rebalanced sample is 50% clean against the pilot's 13% and clean items are easier to transcribe. It replicates. The pooled rise from 0.788 to 0.835 sits well inside the pilot's interval and should not be read as an improvement.

Two things worth noting. First, the composition effect this stratification was registered to guard against *did not materialize*: the two strata came out nearly identical (0.830 vs. 0.839), so pooling would have been safe after all. The guard was still correct to register in advance – that it proved unnecessary is only knowable afterwards. Second, the pilot's clean-stratum estimate of 0.940 was noise from 13 items, and settles at 0.839 with 150.

Sensitivity holds at the larger n (`auroc_sensitivity`):

Subset	n	AUROC	95% CI
Full	300	0.835	$[0.787, 0.879]$
Excluding all parse failures	255	0.839	$[0.790, 0.886]$
Excluding max-entropy items	239	0.794	$[0.737, 0.848]$
Excluding both	206	**0.796**	$[0.736, 0.852]$

5.3 Reasoning: the model cannot do the task (failed)

Metric	Value	
Grading accuracy, balanced sample	**0.517**	(baseline 0.500)
on error-containing answers ($n = 150$)	0.947	
on clean answers ($n = 150$)	**0.087**	
Answers "there is an error"	**93%** of items	
Reasoning entropy AUROC, pooled	0.522 $[0.462, 0.582]$	**no signal**
Items at zero entropy (all 5 samples agree)	126/300 (42%)	
Distinct entropy values occurring	3 in practice	

This is the central result of Phase 2 and it is unambiguous. **With the class imbalance removed, the model grades at chance.** Its Phase 1 accuracy of 75% was not competence; it was a strong prior toward "error" meeting a test set that was 87% errors. Given a balanced set, that prior is exposed: it identifies 94.7% of genuine errors and 8.7% of clean answers, which is close to the behaviour of a constant predictor.

Reasoning entropy therefore has nothing to measure. Its distribution is nearly degenerate – 42% of items have all five samples agreeing, and only three entropy values occur at all – and its AUROC is 0.522 with a CI comfortably containing chance.

The correct reading is not that the entropy metric is poorly implemented. It is that **sampling entropy is a measure of self-consistency, and self-consistency only predicts correctness when the model has competence for the disagreement to track.** A confidently wrong model is stable and wrong; entropy sees the stability and infers reliability. This is a property of the technique, not a bug in this implementation, and it applies to any sampling-based uncertainty method.

Phase 4 (§7) revisits this claim with a larger model. The pooled failure described here turns out to be specific to pooling an error-biased model's predictions, not a general statement that reasoning entropy has nothing to measure: within the `has_error`=1 stratum, a 7B model's entropy becomes a confirmed, sizable predictor once given enough misgraded items to measure against.

5.4 The pre-registered stratified hypothesis was untestable

Phase 1's most promising reasoning claim was that entropy scores 0.756 within the `has_error`=1 stratum and is only cancelled by pooling. Phase 2 registered a threshold of 0.70 for it. The outcome:

Stratum	AUROC (95% CI)	n_{wrong}	Verdict
Pooled	$0.522\ [0.462, 0.582]$	145	**no_signal**
`has_error`=1	$0.836\ [0.626, 0.946]$	**8**	**inconclusive_underpowered**
Clean	$0.239\ [0.128, 0.374]$	137	**inconclusive_underpowered**

The power calculation behind the $n = 300$ choice assumed grading errors would spread across both strata. They did not: the model's bias concentrated almost all of them in the clean stratum, leaving only 8 misgraded items among 150 error-containing ones. That is below the registered minimum of 30, so the honest verdict is *untestable*, not 0.836. Reporting the point estimate here would repeat exactly the mistake the pre-registration existed to prevent.

The clean stratum is inverted as predicted (0.239, with the interval excluding 0.5 by a wide margin), which supports the confident-wrongness account of §4.5. But it has only 13 correct items, so it too falls under the registered minimum. *This verdict changed after the run*: the inversion branch of `classify_scaleup_result` was initially skipping the minority-class guard that every other branch applied – a coding inconsistency, not a registered choice. Applying it uniformly moved this from "confirmed" to underpowered. The threshold was not altered; the fix is recorded here because it was made after seeing the data.

5.5 Replications and integrity

Phase 2 reproduces the pilot's incidental findings, which is reassuring about the pipeline:

Check	$n = 100$	$n = 300$
Digit contradicts its own reasoning	12%	11.0%
Transcription parse-failure rate	4.0%	3.6%
Grading parse-failure rate	1.0%	0.9%
Temperature-0 anchor non-zero	0/100	0/50
Max-entropy abstention rule precision	$14/14 = 1.000$	$\mathbf{57/61 = 0.934}$

The abstention rule is the one place where a Phase 1 number was optimistic: perfect precision on 14 items becomes 93.4% on 61. That is the expected direction for a small-sample extreme, and 93.4% remains a usable operating point – it flags 57 of the 182 transcription errors (31% recall) with 4 false positives.

5.6 The deferral rule: risk-coverage, AURC, and a conformal guarantee

The abstention rule above is one hand-picked operating point. The general version of the same idea is a *risk-coverage curve*: order items by entropy, most confident first, and ask what error rate results from answering only the most confident $X\%$ and deferring the rest.

Coverage	Items answered	Items deferred	Risk (error rate answered)
12.7%	38	262	**0.079**
33.3%	100	200	0.260
52.0%	156	144	0.372
100% (answer everything)	300	0	0.607

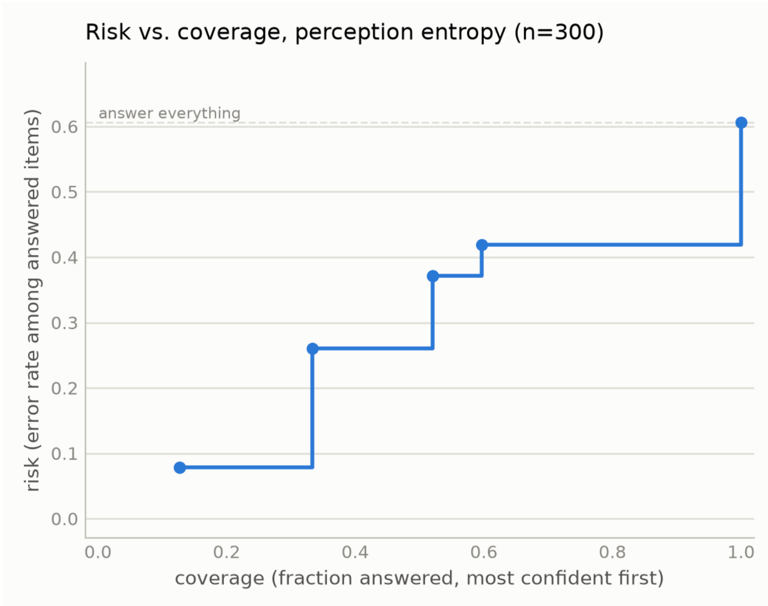

Figure 1: Risk-coverage curve for perception entropy, $n = 300$. A step plot, not a smooth curve, because $K = 5$ entropy has only 7 achievable operating points – the curve is genuinely flat between them and jumps at each one. Coverage 1.0 (answer everything) recovers the overall error rate, 0.607.

A single number summarizes the whole curve: the *area under the risk-coverage curve* (AURC), the average risk swept across all achievable coverage levels. Lower is better, and it must always be read against the random-deferral baseline (just the overall error rate) since a high base error rate caps how good any AURC can be regardless of how well entropy ranks errors.

AURC (entropy-ordered deferral)	**0.410**
Baseline AURC (random deferral order)	0.607
Improvement	0.197

Does the guarantee hold on data it has not seen? A hand-picked cutoff carries no guarantee – it is only known to have worked on the data it was chosen from. A *conformal* threshold is calibrated on one half of the items and evaluated on the other half, repeated over 1,000 random splits, to check whether it actually controls risk on held-out data. Targeting at most 30% risk:

Target risk	30%
Mean coverage achieved	12.6%
Mean risk on held-out data	**13.7%**
Splits where the target was violated	2.4%
Splits where no threshold could reach the target at all	**66.7%**

The guarantee holds – observed risk (13.7%) sits well under the 30% target, and it is violated in only 2.4% of splits, both consistent with a conservative, working conformal procedure. The honest caveat is the fourth row: two-thirds of splits could not even attempt the target, because $K = 5$'s seven operating points do not include one fine enough. This is the same resolution limit noted in

§6.1's $K = 10$ result, stated here in the specific terms that matter for deployment – more samples per item would not just raise the headline AUROC, it would unlock finer-grained, more reliable deferral thresholds. `pilot.plotting.risk_coverage_curve`, `aurc`, and `evaluate_conformal_abstention` implement all of this and are unit tested against both synthetic cases and this exact $n = 300$ run.

6 Phase 3: Closing the Baseline Gap and Testing Prompt Fixes

Phase 2 left two things unaddressed: perception entropy had never been compared against an independent uncertainty signal (only against a cruder version of itself, a plain count of distinct answers), and the reasoning arm's 93% "there is a mistake" response rate had never been tested against an alternative prompt. Both were run in one session (`pilot/04_confidence_and_prompts.ipynb`), reusing the Phase 2 sample so results join directly onto it.

6.1 Entropy vs. token confidence: entropy wins decisively (resolved)

Capturing per-token log-probabilities requires `output_scores=True`, which forces the model off its fused attention kernel and onto eager attention throughout, including the vision tower – a 14 GiB allocation that a 16 GiB GPU cannot provide regardless of sampling batch size. This experiment therefore needed a ≥24 GiB GPU (run on an A100); the notebook checks available VRAM and skips the capture on smaller hardware rather than failing.

Five additional transcription samples per item were drawn with scores captured, summarized per sample by the mean transition log-probability (overall fluency) and the minimum (the single least confident token):

Signal	AUROC	95% CI
Entropy ($K = 5$)	**0.835**	$[0.787, 0.879]$
−mean log-probability	0.537	$[0.469, 0.605]$
−min log-probability	0.460	$[0.392, 0.527]$
Paired (entropy − mean log-prob.): +0.297, 95% CI [+0.214, +0.380], **resolved**		

This closes the last open gap in the perception result: entropy is not simply recovering information the model's own next-token probabilities already expose. Mean log-probability barely clears chance on its own and loses to entropy by a wide, resolved margin.

The minimum log-probability figure needs a caveat rather than a clean headline. Its CI sits at or below 0.5, which invites the reading "token confidence is actively misleading" – but 81% of items cluster in a narrow band (−min-logprob > 20.5, i.e. probability $\sim e^{-20}$) that plausibly reflects an artifact of batched generation (padding positions in the shorter sequences of a batch) rather than genuine uncertainty about real content. Splitting on that band does not resolve cleanly either: both halves land inside wide, chance-overlapping intervals (0.467 $[0.393, 0.543]$ on the clustered majority, 0.529 $[0.378, 0.682]$ on the rest). The honest conclusion is that −min-logprob is not a usable signal, not confidently *why* it fails – mean log-probability is the trustworthy half of this comparison, and it alone is sufficient to answer the question this section asks. One methodological note for reuse: these token-confidence values come from a *different* draw of 5 samples than the reference $K = 5$ entropy (the original samples predate logprob capture), not the identical generations – a fair comparison, since both are independent draws from the same model and image, but not a same-output rescoring.

6.2 $K = 10$ modestly sharpens perception, as predicted (resolved)

The same generation pass extends transcription to $K = 10$. Scored against the $K = 10$-recomputed correctness label – the fair comparison, since scoring $K = 10$ entropy against the stale $K = 5$ label answers a different and less meaningful question – the improvement predicted from the Phase 2 leave-one-out subsample check materializes:

	$K = 5$	$K = 10$	Paired difference
AUROC	0.761 $[0.706, 0.813]$	**0.806** $[0.756, 0.854]$	+0.045 $[+0.005, +0.087]$, **resolved**
Distinct entropy values	7	34	

A side effect worth flagging: the "ground truth" majority-vote correctness label is itself not fully stable at $K = 5$ – recomputing the majority with 10 samples flips a handful of items. Correctness labels for this pipeline should be read as an estimate that sharpens with K, not a fixed target.

6.3 No grading prompt variant fixes the yes-bias (screened out)

Three alternative grading prompts were screened against the FERMAT baseline on the first 100 items – a screen, not a full measurement, per a pre-registered rule that only a variant clearing the baseline by a wide margin earns a full 300-item run:

Variant	Accuracy	Answers "there is a mistake"	vs. baseline
Baseline (FERMAT)	0.510	94%	–
Restate result first	0.550	38%	below bar
"Half are correct" framing	0.540	83%	below bar
Elicit 0–100 confidence	0.510	96%	below bar

None clears the bar. The *restate* variant is the interesting negative: it collapses the model's default answer from 94% "mistake" to 38% – a large behavioral shift – while accuracy barely moves ($0.510 \rightarrow 0.550$). The model is not stuck on a biased default that a better prompt frees it from; it simply swaps which near-constant answer it gives. This weighs against "yes-bias is a fixable prompt artifact" and for a genuine capability ceiling on this 3B model, strengthening the case for testing a larger model (§5.3) as the next real lever on the reasoning arm, rather than further prompt engineering.

7 Phase 4: A Larger Model, and the Reasoning Arm's Stratified Resolution

> **RETRACTED INTERPRETATION.** The `has_error`=1 stratified reasoning results in this section are arithmetically correct and their interpretation is withdrawn — within a single-truth-label stratum, correctness is the same column as the model's own verdict, so the AUROC is near-circular and a signal-free biased coin scores higher. See §. The perception results in this section are unaffected.

Phase 3 left model capability as the one remaining lever on the reasoning arm (§6.3). This phase runs that check on Qwen2.5-VL-7B, finds it ambiguous rather than clearly positive or negative, follows the stratified pattern Phase 2 flagged but could not test, and then closes the power gap that made that pattern untestable.

7.1 The capability gate: marginal, not resolved either way

Metric	3B ($n = 300$, §5.3)	7B ($n = 300$)
Grading accuracy	0.517	**0.590**
vs. baseline	0.500	0.500
Gate verdict (0.55 / 0.65 bars)	**at chance**	**marginal**
Answers "there is an error"	93%	80%

The gate exists precisely so a middling result is not silently read as a resolution. 7B improves 9 points over baseline – a real gain over 3B's 0.517 – but does not clear the pre-registered 0.65 bar that would treat the entropy numbers below as validated rather than diagnostic (`entropy_result_meaningful` is `False` for this run). Consistent with that, the pooled AUROC is 0.520 $[0.458, 0.582]$ – statistically indistinguishable from 3B's 0.522 – and is reported here for completeness, not as evidence on its own.

7.2 The stratified pattern replicates, initially still underpowered

Stratum	AUROC (95% CI)	n_{wrong}	Verdict
Pooled	0.520 $[0.458, 0.582]$	123	**no signal**
`has_error`=1	0.761 $[0.646, 0.860]$	17	**underpowered**
Clean	0.280 $[0.200, 0.366]$	44	**confirmed inversion**

Both `sign_reversal` and `pooled_understates` hold, same as 3B. Two comparisons are worth making against 3B's own $n = 300$ result (§5.4), not the earlier $n = 100$ pilot number, since only the $n = 300$ runs share a design:

- `has_error=1`: 3B scored 0.836 $[0.626, 0.946]$ on just **8** misgraded items – too few to trust on its own. 7B's 0.761 here is still underpowered too (17 misgraded), but sits inside 3B's wide interval. The two independent point estimates agreeing this closely is a hint the underlying effect size is real and roughly this large, even though neither alone clears the power bar.
- **Clean stratum**: 3B's clean stratum was underpowered for a *different* reason – not too few items, but too few *correct* ones (13 of 150, because 3B is wrong on clean items 91% of the time). 7B's clean-item accuracy is far better (106 of 150 correct), which is what makes 44 misgraded items available to measure against – the first clean stratum in this project to clear the power bar. The confirmed result, 0.280 $[0.200, 0.366]$, is close to 3B's own point estimate of 0.239 $[0.128, 0.374]$: the direction and rough magnitude were right at 3B, just unmeasurable there.

7.3 Manual inspection: four mechanisms, no label noise

Before spending more compute, the 17 misgraded `has_error`=1 items from the reference run were read directly. This required a pipeline change: the per-item checkpoint always retained the model's full raw text, but the final scoring step had been discarding it down to digit, entropy, and correctness – adequate for computing AUROC, not for explaining a wrong answer. Carrying the raw text through to the final CSV made the following possible.

Two mechanisms account for most of the 17:

- **Digit/reasoning self-contradiction**, elevated relative to 3B. Roughly 35–40% of digit-0 samples among the wrong items have reasoning text that itself raises a concern, after correcting a naive keyword check's false positives (23% of raw keyword matches were phrases like "no error", which trips a literal string match on "error" without asserting one). This is *not* a controlled comparison against 3B's previously reported 11% (§5.3), since 3B's samples were not re-measured with the same correction – reported as a directional observation, not a resolved gap.
- **Pattern-matching a solution's structure instead of independently recomputing a step.** For example, one item's worked solution states $19 \times (-18) \times 23 = -7966$; the model's reasoning restates this product as confirmation without recomputing it. The correct value is -7866.

Of the 17, **2 were unanimous across all $K = 5$ samples** (reasoning entropy exactly 0) and showed no error anywhere in the pert_a *text* – initially read as possible dataset label noise. Checking both directly against the source handwritten images (`pilot/07_manual_case_inspection.ipynb`) instead overturned that reading: both have real errors that the text excerpt had simply failed to surface. One item's final line reads, verbatim, "The required number is y" – the algebra is correct ($y = 7$) but the answer is never actually substituted in. The other defines $\mathbb{R} \times \mathbb{R} \times \mathbb{R}$ as $\{(x, y, z) : x, y, x \in \mathbb{R}\}$ – the tuple introduces z but the membership condition repeats x instead, an internal inconsistency in the model's own reference solution. Re-running the same check after §7.4's 200-item extension found a third unanimous case among the new items, with the same pattern: correct arithmetic ($\sqrt{45} = 3\sqrt{5}$ for a 3D distance calculation) followed by an unwarranted final claim – the answer is labeled "meters," a unit the problem never specified for two bare coordinate points.

All three unanimous cases have real, findable errors – none are label noise – and together they sharpen the two mechanisms above into four. The common thread across the new two: the model appears to verify the *core computation* of a solution and stop there, without checking whether the final answer statement is actually complete (an unsubstituted variable), internally consistent (mismatched notation), or licensed by the problem (a unit that was never given). This is a more specific description of the capability ceiling than "cannot do arithmetic" – the arithmetic itself is not what these three items get wrong.

7.4 Closing the power gap: `has_error=1` confirmed

200 additional `has_error=1` items were drawn – disjoint from the reference run by construction, since both draws index the same seeded shuffle of error items and take non-overlapping slices (`pilot.data.load_fermat_extra_error_items`) – and merged with the 300-item reference run, giving 350 `has_error=1` items against 150 clean ones (500 total, deliberately no longer balanced: the extension targets only the stratum that needed power).

Stratum	AUROC (95% CI)	n_{wrong}	Verdict
`has_error=1` (before, $n = 150$)	0.761 [0.646, 0.860]	17	**underpowered**
`has_error=1` (after, $n = 350$)	**0.834** [0.768, 0.891]	**38**	**confirmed**
Clean (unchanged, $n = 150$)	0.280 [0.200, 0.366]	44	**confirmed inversion**

n_{wrong} clears the registered minimum of 30 for the first time in this stratum, and the AUROC itself rose from 0.761 to 0.834, clearing the pre-registered 0.70 threshold (§5.4) with a CI that excludes chance outright. This is the first point in the project where *both* grading strata are simultaneously powered and resolved, in opposite directions – not a hypothesis about pooling hiding a real effect, but a fully measured, symmetric result.

One trap in this specific sample is worth flagging explicitly. Its pooled AUROC reads 0.612 $[0.560, 0.663]$ – above chance, which invites the reading that the sign-reversal problem has resolved itself. It has not: the sample is no longer 50/50 on `has_error` (350 vs. 150), which mechanically pulls the pooled statistic toward the larger, stronger `has_error`=1 stratum. `stratified_auroc` remains the only correct read of this data, exactly as it was when pooling produced the opposite distortion in §5.3.

Net revision to the reasoning-arm conclusion: §5.3's finding stands as stated – pooled entropy on an error-biased 3B model measures nothing. But that is now known to be a statement about pooling an error-biased model's output, not about reasoning entropy in general. Within the `has_error`=1 stratum, at 7B, entropy is a confirmed AUROC-0.834 predictor of grading correctness, and its close agreement with 3B's own (underpowered) 0.836 point estimate for the same stratum suggested this was close to the true effect size at 3B too – just never measurable there. §7.5 tests that suggestion directly rather than leaving it as a hint. The practical implication is not "bigger model, better entropy" so much as "stratify by the model's own response bias before pooling, or a real signal disappears into it."

7.5 3B confirms the same result: the effect is model-size-independent

The previous section's agreement between 7B's confirmed 0.834 and 3B's underpowered 0.836 was suggestive, not established – two numbers can land close together by chance when one of them has a wide interval. The direct test is to power 3B's own `has_error`=1 stratum the same way 7B's was: 500 additional `has_error`=1 items, drawn and merged exactly as in §7.4. **500, not 200**: 3B's error rate on this stratum ($8/150 = 5.3\%$) is roughly half 7B's ($17/150 = 11.3\%$), since 3B answers "there is an error" even more often (93% vs. 80%) and so gets more `has_error`=1 items right by default, leaving fewer wrong ones to power against.

	AUROC (95% CI)	n_{wrong}	Verdict
3B, before ($n = 150$)	0.836 $[0.626, 0.946]$	8	**underpowered**
3B, after ($n = 650$)	**0.854** $[0.796, 0.902]$	**42**	**confirmed**
7B, after ($n = 350$, §7.4)	0.834 $[0.768, 0.891]$	38	**confirmed**

Both model sizes now clear the registered power bar and the 0.70 threshold independently, landing within 0.02 of each other. **This resolves the open question directly: the `has_error=1` stratified effect is model-size-independent, not a property specific to 7B.** The clean stratum was not re-tested at 3B (the extension targets only `has_error`=1 items by design) and remains at its original, underpowered 0.239 $[0.128, 0.374]$ on 13 correct items – unlike 7B's clean stratum, 3B's has never cleared the power bar, because 3B is wrong on clean items far more often (91% vs. 7B's 29%), which is itself consistent with 3B's stronger "there is an error" bias.

The same pooled-AUROC distortion documented in §7.4 recurs here and is worth re-flagging rather than assuming the earlier caveat covers it: this CSV's naive pooled AUROC reads 0.610 $[0.565, 0.656]$, again above chance, again purely because the sample is no longer 50/50 (650 `has_error`=1 vs. 150 clean). `stratified_auroc` is the only correct read of both the 7B and 3B stratum-powered datasets.

With this result, the reasoning arm's revised conclusion (§7.4) is no longer contingent on 7B specifically: **entropy predicts grading correctness within the `has_error=1` stratum at**

both model sizes tested, confirmed and in close agreement. What differs by model size is not whether the signal exists but how error-biased the model is, which determines how many items it takes to measure the signal, not whether the signal is there to measure.

7.6 Matching 7B to the same 650 items: still confirmed, a somewhat wider gap

The comparison above used 7B at $n = 350$ against 3B at $n = 650$ – different totals, because 7B reached the power bar with fewer items. As a check that the different totals were not themselves doing any work, 7B's stratum was extended a second time (300 more `has_error`=1 items, `skip=350`) to reach the same 650-item total 3B was tested on, drawn from the identical seed=42 ordering. (2 of the freshly drawn items turned out to duplicate items already in the first extension – FERMAT's Hub copy appears to have shifted slightly between the two draws – and were dropped after inspection confirmed they were ordinary items, not a sampling bug; the matched total is 648, not 650.)

	AUROC (95% CI)	n_{wrong}	Verdict
7B, $n = 350$ (§7.4)	0.834 [0.768, 0.891]	38	**confirmed**
7B, $n = 648$ (matched to 3B)	**0.801** [0.751, 0.846]	**73**	**confirmed**
3B, $n = 650$ (§7.5)	0.854 [0.796, 0.902]	42	**confirmed**

The matched-n result is still confirmed – comfortably above the 0.70 threshold, CI excluding chance – and 0.801 sits inside the original $n = 350$ result's own interval, so this is a refinement from more data, not a contradiction. It is a somewhat less striking match to 3B's 0.854 than the earlier $n = 350$-vs-$n = 650$ comparison's 0.834 suggested: the two CIs still overlap ($0.846 > 0.796$), so the two model sizes remain statistically indistinguishable, but the point estimates are 0.05 apart rather than 0.02. **The conclusion is unchanged – both model sizes confirm the `has_error`=1 effect, and it is not resolvably different between them – but "within 0.02" overstated how tight that agreement is once both are measured on the same items.** Reported here rather than quietly revised in place, since the closer-looking number was the one already circulated before this check ran.

8 Phase 5: A Second Model Family (LLaVA-NeXT)

> **RETRACTED INTERPRETATION.** The `has_error`=1 stratified reasoning results in this section are arithmetically correct and their interpretation is withdrawn — within a single-truth-label stratum, correctness is the same column as the model's own verdict, so the AUROC is near-circular and a signal-free biased coin scores higher. See §. The perception results in this section are unaffected.

Every result so far is one model family, Qwen2.5-VL, at two sizes. The single largest remaining external-validity gap is whether either result is Qwen-specific. This phase repeats both headline measurements – perception AUROC and the `has_error`=1 stratified reasoning AUROC – on **LLaVA-NeXT** (`llava-hf/llava-v1.6-mistral-7b-hf`, ∼7B, a different architecture and training recipe entirely) at the identical $n = 300$ balanced sample every reference run in this report used.

8.1 Perception: gated by a real capability gap, not usable

Transcription accuracy is 3.0% (9/300), against Qwen's ∼42%. This was diagnosed directly rather than reported at face value. Two checks ruled out a pipeline bug: independently recomputing `transcription_correct` from the raw text matches the stored value on all 300/300 rows (not a scoring bug), and the parse-failure rate is only 9.6% (144/1500), so the model is not failing to follow the `**Question:**`/`**Answer:**` format either. The actual cause, found by reading real samples and then confirming it systematically:

Item type	n	Accuracy
Multiple-choice	57	12.3%
Free-response	243	**0.8%** (2 correct)

LLaVA-NeXT can occasionally identify a multiple-choice letter but is at a practical floor on free-response transcription, which is 81% of the dataset. This is the same capability-gate logic as §7.1 applied to the perception arm for the first time: an AUROC computed from 9 correct items (0.710 $[0.525, 0.886]$) is an order of magnitude below the registered power minimum used everywhere else in this report and is not reported as a result. **Perception's entropy claim requires a second model with real OCR competence to test** – not yet chosen.

8.2 Reasoning: confirmed, the first cross-model-family replication in this project

The reasoning arm does not depend on fine-grained OCR – grading is a holistic judgment from the image – and is unaffected by §8.1's gate. The $n = 300$ reference run found a promising but underpowered echo of the Qwen results: clean-stratum 0.283 $[0.178, 0.398]$ (14 correct items), `has_error`=1 stratum 0.766 $[0.628, 0.883]$ (14 misgraded items) – both short of the registered minimum of 30. 250 additional `has_error`=1 items (sized from the observed 9.3% error rate on this stratum, 90% Wilson CI [6.1%, 14.0%]) were drawn and merged, mirroring §7.4/§7.5 exactly:

	AUROC (95% CI)	n_{wrong}	Verdict
LLaVA, before ($n = 150$)	0.766 $[0.628, 0.883]$	14	**underpowered**
LLaVA, after ($n = 400$)	**0.775** $[0.694, 0.848]$	**34**	**confirmed**
Qwen-7B ($n = 648$, §7.6)	0.801 $[0.751, 0.846]$	73	**confirmed**

LLaVA-NeXT's `has_error=1` stratum clears the registered power bar ($34 \geq 30$) and the 0.70 threshold, with a CI (upper bound 0.848) that overlaps Qwen-7B's confirmed interval almost entirely. The two model families are not resolvably different on this measurement. The clean stratum remains at its original, underpowered 0.283 (14 correct items – the extension targeted only `has_error`=1) but is itself close to Qwen-7B's confirmed 0.280, and the same response bias that drives the sign-reversal pattern replicates too: LLaVA answers "there is an error" 91.1% of the time, close to Qwen-3B's 93% and higher than Qwen-7B's 80%.

This is the first confirmed replication across model families in this project. Every prior confirmation (§7.4, §7.5, §7.6) compared 3B against 7B within Qwen2.5-VL. A second, architecturally distinct model reproducing the same stratified effect at a statistically indistinguishable magnitude is evidence the effect is a property of this task and pipeline, not of Qwen specifically – directly addressing the single largest gap identified for this line of work.

9 Phase 6: A Third Model Family (InternVL3), and a Sensitivity-Cut Failure

Phase 5 answered "is either result Qwen-specific" for the reasoning arm but left perception open, gated by LLaVA-NeXT's near-zero free-response OCR. This phase repeats both headline measurements on **InternVL3-8B** (`OpenGVLab/InternVL3-8B-hf`, trained independently of both Qwen2.5-VL and LLaVA-NeXT's base LLM), at the identical $n = 300$ balanced sample. The first integration attempt used InternVL3's own `trust_remote_code` model card recipe and crashed on a real run (`transformers` v5 requires custom model classes to initialize `all_tied_weights_keys` via `post_init()`, which that recipe's code predates); the checkpoint used here is `transformers`' native integration instead, which restores standard batched `generate()` sampling the `trust_remote_code .chat()` interface never had.

9.1 Perception: clears the accuracy bar, fails the sensitivity cut

Transcription accuracy is 21.0% (63/300) – about half Qwen's ∼42%, but, unlike LLaVA-NeXT's 0.8%, well above the registered 30-item power minimum, so this is not a capability gate the way §8.1 was. The raw perception AUROC is **0.915** $[0.880, 0.944]$ – higher than Qwen's confirmed 0.835, despite the lower base accuracy. That mismatch does not happen with a genuine graded uncertainty signal, so it was checked the way every surprising result in this report is checked before being reported.

Every derived column recomputes from the raw per-item samples (the same `parse_transcription` → `canonical_answer_label` → `cluster_entropy` sequence the scoring cell uses) with **zero mismatches against the stored CSV** – this is real model behavior, not a pipeline bug. But it does not survive the standard robustness check (`auroc_sensitivity`, §4.5) every other perception number in this report is held to:

	AUROC (95% CI)	Verdict
Full ($n = 300$)	0.915 $[0.880, 0.944]$	–
Excluding max-entropy items ($n = 98$)	**0.556** $[0.495, 0.614]$	**chance**
Qwen-3B, same cut (for comparison)	0.796 $[0.736, 0.852]$	**survives**

The mechanism, found by categorizing every one of the 300 items rather than eyeballing a handful: 202/300 items (67%) pin at exact maximum entropy ln(5), and within that group accuracy is a **0.99% floor (2/202)**; among the other 98 items accuracy is **62.2% (61/98)**. This is a bimodal coherent-vs-incoherent split driving the whole headline number, not a graded relationship between disagreement level and correctness. An initial reading (based on 1–2 examples showing a degenerate repeated-phrase failure, e.g. the same clause repeated dozens of times) attributed this to generation loops specifically; a full-population check found that pattern in only ∼30% of the max-entropy-wrong group (61/200). The dominant mechanism (∼70%, 139/200, confirmed by reading concrete examples) is different: InternVL3's five retries each land on a different *incomplete slice* of a multi-step derivation – one restates a garbled version of the question, another jumps into an intermediate algebra step, another states an unfinished fragment as its "Answer" – rather than reliably converging on a stated final result the way Qwen typically does even when wrong. Every one of these fields is non-empty and canonicalizes to a real (if messy) label, so the parser is behaving correctly on genuinely incomparable content, not failing.

This result must not be reported as "InternVL3 beats Qwen on perception." It is the clearest illustration in this report of why the sensitivity cut exists: a headline AUROC can be manufactured by an extreme bimodal failure mode that has nothing to do with a model being usefully, gradedly uncertain.

9.2 Reasoning: powered, resolved above chance, still short of confirmation

The reasoning-entropy distribution shows no equivalent pinning problem (4 reachable values, no single value above 54% of items), so this stratum's result is not an artifact of the same kind. Stratifying by `has_error`:

	AUROC (95% CI)	n_{wrong}	Verdict
`has_error`=1	0.628 [0.548, 0.706]	45	**resolved, below threshold**
`has_error`=0 (clean)	0.369 [0.290, 0.454]	97	**inverted, as predicted**

Both strata clear the registered $n = 30$ power minimum. The clean stratum replicates every other model/family's inversion. The `has_error`=1 stratum is a genuinely new outcome for this report: its CI excludes chance (lower bound 0.548) but its upper bound (0.706) sits just under the 0.70 confirmation threshold used for Qwen-3B/7B and LLaVA-NeXT – powered and resolved above chance, but not **confirmed** and not **inconclusive_underpowered** either. Reading the model's own grading text on wrong-vote items surfaces a plausible reason this stratum is weaker here than elsewhere: on at least one case the model's own stated reasoning defaults to "no error" explicitly because it judges the image unreadable (*"the image contains repetitive text... no clear answer given, and therefore no error can be evaluated"*) – the same generation instability documented in §9.1 leaking into the reasoning arm's inputs, not a separate mechanism.

9.3 A pre-registered screen of three fixes: none resolves

Rather than iterating on fixes until one cleared 0.70 – which, with enough attempts, happens by chance alone – three ideas were screened once each on the same 150 `has_error`=1 items, with the pass bar (AUROC $\geq$ 0.70, CI excludes chance, minority class $\geq$ 30) written before the run. This mirrors how the Phase 3 grading-prompt screen was run and reported (§6.3), where "none clear the bar" was itself the finding. The three: an explicit "do not default to no-error on illegible input" instruction (`commit`, motivated by the wrong-vote case quoted above); a retest of Phase 3's restate-result-first variant, unmodified; and a $K = 10$ resample, motivated by the CI upper bound sitting just under the threshold.

	AUROC (95% CI)	Paired diff vs. baseline	Resolved?
Baseline ($K = 5$)	0.628 [0.548, 0.706]	–	–
`commit`	0.698 [0.618, 0.773]	+0.069 [−0.028, +0.166]	**no**
`k10`	0.702 [0.616, 0.781]	+0.074 [−0.024, +0.171]	**no**
`restate`	0.621 [0.541, 0.699]	−0.008 [−0.112, +0.095]	**no**

No variant resolvably beat the baseline: all three paired intervals span zero. Reported honestly, `k10` does meet the bar as written (0.7023, CI excludes chance, 34 misgraded items) – but it clears the threshold by 0.0023 while its own interval runs [0.616, 0.781], so the lower bound sits far below the bar it passed, and the 0.005 separating it from `commit` (which "fails") is noise. This exposes a real limitation in the pre-registered bar itself: 0.70 was borrowed from the confirmation runs of §7–§8, where CIs were tight and unambiguously above it; applied to an $n = 150$ screen with a ±0.08 interval, "the point estimate crossed 0.70" is a much weaker statement. `k10` also changes the target as well as the predictor (21 items flip correctness label at $K = 10$), a confound the paired interval cannot remove. **Decision: no confirmation run; 0.628 stands as InternVL3's reported reasoning result.**

Two secondary results are worth keeping. First, $K = 10$'s effect on *grading accuracy* does resolve where its AUROC does not: $0.700 \to 0.773$, $+0.073$ $[+0.013, +0.133]$, with 16 items fixed against 5 broken – more samples make the majority vote more accurate, the same direction as the confirmed $K = 5 \to K = 10$ perception result (§6.1). Second, `restate` reproduces its Phase 3 failure signature on a different model family entirely: heavy churn (31 items fixed, 36 broken) with accuracy moving *down*, $0.700 \to 0.667$ – the model swapping which near-constant answer it gives rather than discriminating, exactly as on Qwen-3B.

10 Phase 7: A Second Dataset (ScratchMath) — Gated

Every result to this point is on one dataset. With three model families agreeing on the `has_error`=1 effect, a second dataset became the largest remaining external-validity gap. This phase holds the model fixed at Qwen2.5-VL-7B (confirmed 0.801 on FERMAT at $n = 648$) so that only the dataset varies, and runs a sizing check on **ScratchMath** (1,720 handwritten primary/middle-school scratchwork items).

10.1 What the candidate datasets could and could not test

Two benchmarks were viable candidates; both were inspected directly rather than taken from their papers, and both turned out to consist *entirely* of error-containing items — ErrorRadar has 0/2500 rows without an error annotation, and on ScratchMath the student's answer never equals the reference answer (0/1479). Two consequences follow immediately: no balanced sample can be constructed, and **the clean-stratum inversion confirmed in §7 cannot be tested on either**. What remains testable is the `has_error`=1 stratum, which is the cross-family-confirmed result. ErrorRadar was excluded on a second ground: its images are the *problem figures* and the student work is a text field, so there is no handwriting to grade.

Because every item contains an error, the model's standing "there is an error" bias scores well by construction, so grading accuracy is not the quantity of interest. The pre-registered gate was set on the *misgrade rate* — the minority class the AUROC needs — at 5%, before the run.

10.2 Result: gated, and not merely underpowered

Grading accuracy	96.0% *(by construction)*	
Says-error rate	90.0%	
Parse failures	0.0%	
Items at zero entropy	70/100	
Misgraded	**4/100 = 4.0%**	(bar: ≥5%)

The misgrade rate falls under the bar, so the stratum cannot be powered without roughly 750 items. But the decisive evidence is qualitative, and quantified over all 500 samples rather than read off a few: **in 24% of samples (120/500) the model explicitly states it cannot read or use the image** — "the image does not contain any handwritten work", "impossible to determine" — **and in 82% of those (98/120) it answers `Error: 1` regardless**. On an all-error dataset that scores as correct. A large share of the 96% is therefore the response bias firing through explicit non-engagement, not grading competence.

One further check settles whether the non-engagement could simply be filtered out. It cannot: discarding every item on which the model states it cannot read the image leaves 40 of 100 items, and **all four misgraded items are among those discarded**. The retained subset therefore has no minority class at all, and the AUROC is not merely weak but undefined (`nan`). **Removing the outputs where the model says it cannot read the image removes every misgraded case, leaving no minority class; the analysis becomes undefined rather than cleaner.** The non-engagement cannot be separated from the measurement here, which is the structural form of the same conclusion. Note also that the partition is not clean — only 3 of 100 items have all five readings flagged, the rest are mixed — and that the phrase list used to detect non-engagement was written after reading the failures, so this cut is admissible as evidence that the setting cannot be measured but would not support a positive claim.

This is why the verdict is "not comparable" rather than "underpowered". Non-engagement is concentrated on the misgraded items (0.40 of samples versus 0.23 on correctly-graded ones), so a powered run would substantially be measuring whether the model can read the scratchwork — a different construct from the FERMAT result, and one that would not be comparable to it even if it cleared the bar. Combined with 70/100 items at exactly zero entropy, there is nothing left for a ranking metric to order. Decision: no further ScratchMath GPU work.

Two things did replicate qualitatively. Among the four misgrades, two are instances of the mechanism documented in §7.3: the model verified the student's *own* incorrect computation rather than recomputing it (accepting 522×5 where the problem required $522 \times 3 \times 5$, and $50 - 12$ where it required $50 - 28$). The failure mode travels across datasets even where the metric cannot.

11 Integrity Checks

Both entropy signals depend on the sampling and parsing pipeline actually working. Two checks validate that before trusting any entropy number.

Check	Result
Temperature-0 anchor, transcription (2 greedy draws/item, must be ≈ 0)	0/100 non-zero
Temperature-0 anchor, grading (2 greedy draws/item, must be ≈ 0)	0/100 non-zero
Transcription parse-failure rate (before marker-regex fix)	13.4% (67/500)
Transcription parse-failure rate (after marker-regex fix)	4.0% (20/500)
Grading parse-failure rate ($K = 5$)	1.0% (5/500)
Grading parse-failure rate ($K = 15$)	0.9% (14/1500)

The temperature-0 anchor passing at 0/100 confirms the sampling/parsing pipeline is behaving deterministically at temperature 0, as it must. The transcription parse-failure rate was itself a real bug (the extraction regex only matched the exact `**Answer:**` markdown the prompt requested, missing common real-world drift such as LaTeX `\textbf{}` bold, plain text, and a full-width colon); widening the pattern recovered most of the missed samples.

12 Summary

12.1 Final conclusions (Phase 2–5)

Arm	Status	AUROC (95% CI)	Basis
Perception	**Works**	**0.835** $[0.787, 0.879]$	Replicated, robust to all cuts
has_error=1 stratum	**Works**	0.830 $[0.760, 0.893]$	Like-for-like vs. pilot's 0.762
harshest artifact cut	**Works**	0.796 $[0.736, 0.852]$	$n = 206$, CI excludes chance
vs. token confidence	**Wins**	$+0.297$ $[+0.214, +0.380]$	Paired, resolved – §6.1
$K = 10$ vs. $K = 5$	**Improves**	$+0.045$ $[+0.005, +0.087]$	Paired, resolved – confirms p
Reasoning, pooled (3B)	**Fails**	0.522 $[0.462, 0.582]$	Model grades at chance (0.51
prompt variants (screen)	**None fix it**	–	3 tried, none clear bar – §6.3
Reasoning, pooled (7B)	**Fails**	0.520 $[0.458, 0.582]$	Capability gate marginal – §
has_error=1 stratum (7B, $n = 350$)	**Works**	0.834 $[0.768, 0.891]$	Confirmed, $n_{\text{wrong}} = 38$ – §7.
Clean stratum (7B)	**Inverted**	0.280 $[0.200, 0.366]$	Confirmed, $n_{\text{wrong}} = 44$ – §7.
has_error=1 stratum (3B, $n = 650$)	**Works**	**0.854** $[0.796, 0.902]$	Confirmed, $n_{\text{wrong}} = 42$ – §7.
has_error=1 stratum (7B, $n = 648$)	**Works**	0.801 $[0.751, 0.846]$	Confirmed, matched to 3B –
Perception (LLaVA-NeXT)	**Gated**	0.710 $[0.525, 0.886]$	Not reportable, $n_{\text{correct}} = 9$ –
has_error=1 stratum (LLaVA)	**Works**	0.775 $[0.694, 0.848]$	Confirmed, $n_{\text{wrong}} = 34$ – §8.
Clean stratum (LLaVA)	**Inverted**	0.283 $[0.178, 0.398]$	Underpowered, $n_{\text{correct}} = 14$
Perception (InternVL3)	**Not real**	$0.915 \rightarrow 0.556$	Fails sensitivity cut – §9.1
has_error=1 stratum (InternVL3)	**Below bar**	0.628 $[0.548, 0.706]$	Powered, resolved, not confir
prompt/K screen (InternVL3)	**None fix it**	–	3 tried, no paired diff resolve
Clean stratum (InternVL3)	**Inverted**	0.369 $[0.290, 0.454]$	Powered, $n_{\text{wrong}} = 97$ – §9.2
Second dataset (ScratchMath)	**Gated**	–	Not comparable; 82% of non-

The 3B pooled reasoning result is a statement about *that model*, not the metric. See §5.3. Perception now has both an independent baseline it beats and a confirmed gain from additional sampling; the reasoning arm's pooled failure has been shown not to be a simple, promptable artifact (§6), and – once stratified and adequately powered at *both* model sizes – not a statement that the technique has nothing to measure, either (§7). 7B's matched-n result (0.801) and 3B's (0.854) have overlapping CIs – not resolvably different – which is the strongest evidence in this report that the has_error=1 stratified effect is a property of the task and pipeline, not of one particular model, even though the point estimates are not as numerically close as an earlier, differently-sized comparison suggested. The pooled/stratified split is now the central caveat on reading *any* AUROC in this report: pooling can hide as much as it reveals when a model's own response bias is asymmetric.

Phase 5 extends this from model *size* to model *family*. LLaVA-NeXT's confirmed has_error=1 result (0.775) has a CI that overlaps Qwen-7B's confirmed interval almost entirely – the first cross-family replication in this report, not just cross-size. Its perception arm could not be tested at all: a 0.8% free-response transcription accuracy is a capability gate, the same concept §7.1 used for reasoning, now shown to apply to perception too. The reasoning result generalizes across the two model families tested; whether the perception result does remains open, gated on finding a second model with real OCR competence.

Phase 6 (InternVL3) is the first model where perception clears the accuracy bar but the AUROC itself does not survive this report's own robustness standard – 0.915 collapses to 0.556 under the identical cut that leaves Qwen's 0.835 at 0.796. Read together with Phase 5, the perception result now has one clean replication candidate blocked by a capability gate and one blocked by a sensitivity-cut failure; it remains unconfirmed outside Qwen. Its reasoning has_error=1 stratum (0.628) is real (CI excludes chance) but weaker than every other model/family measured (0.775–0.854) and does not clear the confirmation threshold – introducing this report's first "resolved but not confirmed" verdict, distinct from both ends of the scale used everywhere else.

12.2 Phase 1 exploratory results (superseded)

Retained because they document how the Phase 2 design was arrived at, not as standing claims. Every reasoning row here is superseded by §5.3.

Arm	Status	AUROC (95% CI)	Notes
Perception (final)	**Working**	0.788 $[0.697, 0.869]$	Robust to all artifact cuts
Perception (attempt 1)	**Failed**	–	83% pinned at max entropy
Perception (attempt 2)	**Failed**	–	Over-merged, 95% implausible
Reasoning, digit ($K = 5$)	**Unresolved**	0.618 $[0.489, 0.741]$	CI includes chance
Reasoning, digit ($K = 15$)	**Unresolved**	0.665 $[0.514, 0.800]$	No paired gain over $K = 5$
Reasoning, text ($K = 5$)	**Unresolved**	0.702 $[0.579, 0.817]$	Beats chance, not the baseline
Reasoning, text ($K = 15$)	**Failed**	0.520 $[0.378, 0.664]$	Cascade over-merging

Reading that table by point estimate alone reverses two of its conclusions, so the paired comparisons are collected here explicitly. Exactly one of the five differences Phase 1 examined can be resolved at $n = 100$ – and it is a degradation, not a gain:

Comparison	Paired difference (95% CI)	Resolved?
Reasoning text − digit, $K = 5$	$+0.083$ $[-0.064, +0.233]$	No
Reasoning text − digit, $K = 15$	-0.145 $[-0.317, +0.037]$	No
Reasoning digit, $K = 5 - K = 15$	-0.047 $[-0.178, +0.090]$	No
Reasoning text, $K = 5 - K = 15$	$+0.182$ $[+0.037, +0.330]$	**Yes**
Perception − best reasoning arm	$+0.086$ $[-0.073, +0.243]$	No†

† Perception and reasoning predict errors on *different tasks* with different base rates, so this row is descriptive only – a paired interval is not the right instrument for it. Perception's standing rests on its own CI excluding chance and surviving §3.4's cuts, not on beating the reasoning arm.

13 Limitations

- **The 3B reasoning result is about that model, not about entropy in general – and Phase 4 confirms the caveat mattered.** Qwen2.5-VL-3B cannot grade handwritten math above chance, so Phase 2 establishes that pooled entropy fails *in that regime*, not that the technique fails outright. A 7B model's pooled result is statistically identical (0.520 vs. 0.522), but its `has_error`=1 stratum, once adequately powered, is a confirmed 0.834 – so the caveat this bullet used to raise as untested is now resolved, and resolved in the direction it hoped for (§7).

- **Two model sizes and now three model families agree on reasoning's `has_error`=1 direction; perception has not been confirmed on any family besides Qwen.** 3B and 7B (§7.1), LLaVA-NeXT (§8), and InternVL3 (§9, weaker but still resolved above chance) all point the same direction on `has_error`=1 – evidence the effect is not specific to Qwen2.5-VL, though InternVL3's magnitude (0.628) is the first to fall short of this report's confirmation bar. Perception has no cross-family confirmation yet: LLaVA-NeXT could not test it (a capability gate) and InternVL3 could measure it but the result did not survive the sensitivity

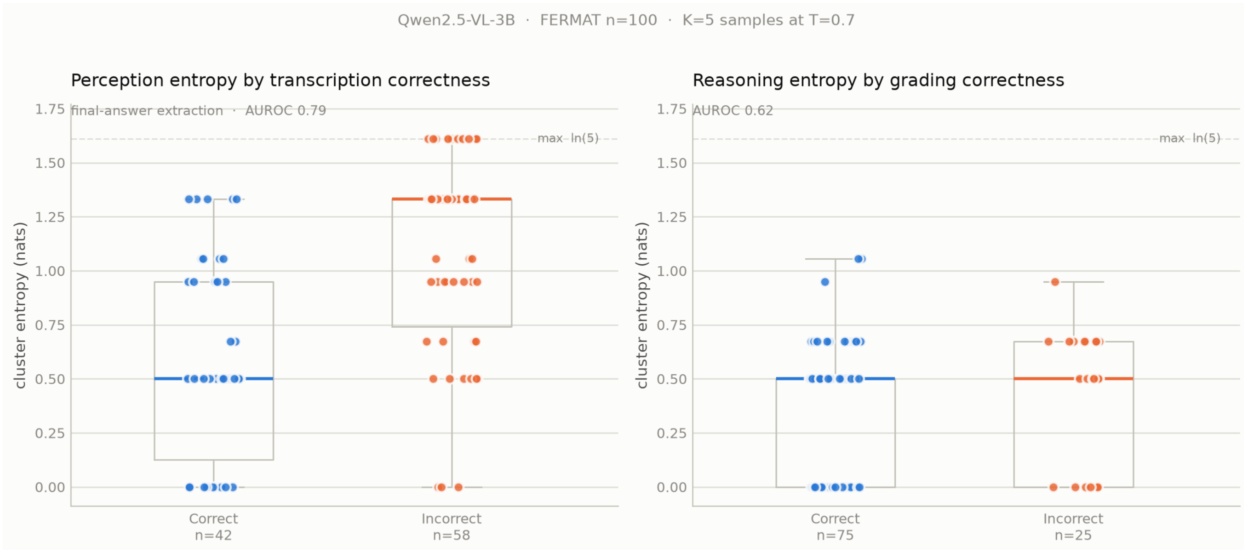

Figure 2: Final result for both arms, $n = 100$. Left: perception entropy by transcription correctness (AUROC 0.788, 95% CI $[0.697, 0.869]$). Right: reasoning entropy by grading correctness at $K = 5$ (AUROC 0.618, 95% CI $[0.489, 0.741]$ – includes chance).

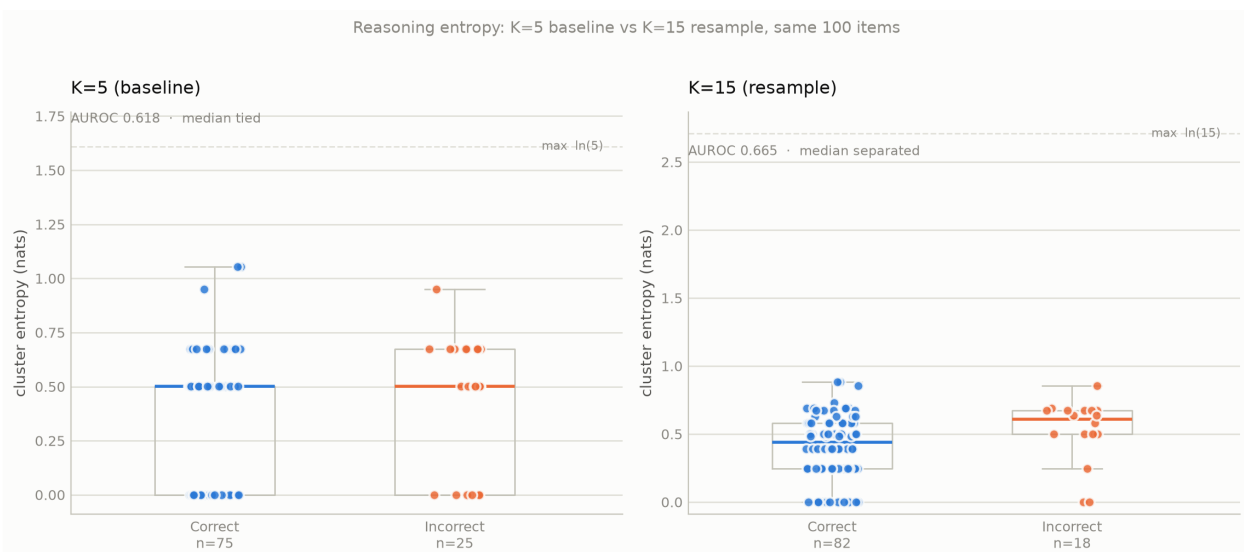

Figure 3: Reasoning entropy before ($K = 5$) and after ($K = 15$) resampling, same 100 items. The exact median tie at $K = 5$ breaks at $K = 15$ – a structural consequence of 13 reachable entropy values instead of 5 – but the paired AUROC difference is -0.047, 95% CI $[-0.178, +0.090]$: no measurable gain in discrimination.

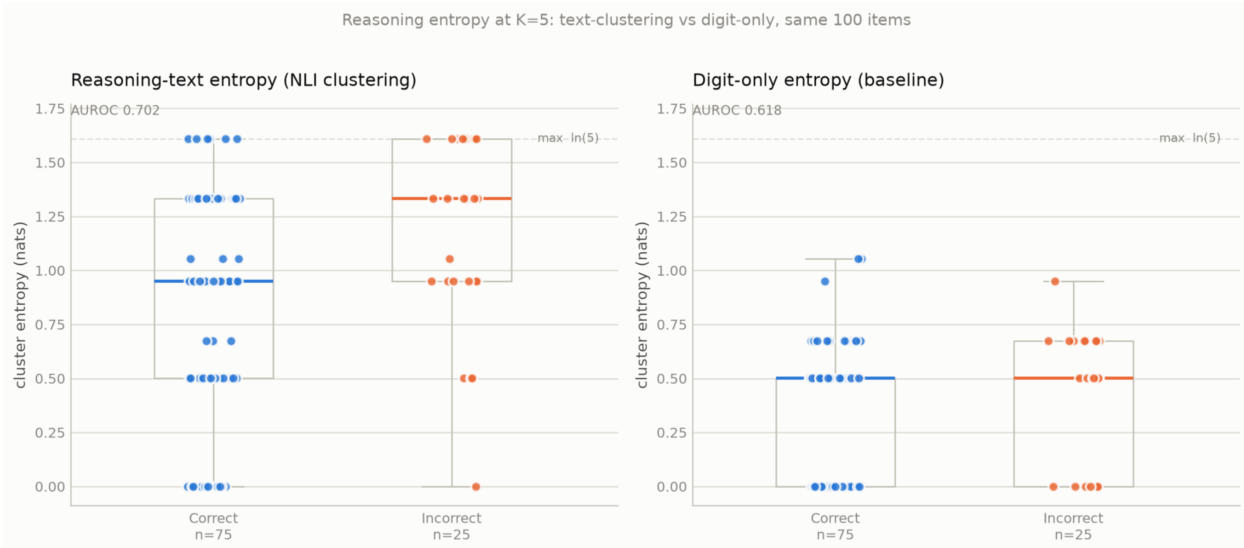

Figure 4: Reasoning entropy at $K = 5$: clustering the model's reasoning text (left, AUROC 0.702) vs. the digit-only baseline (right, AUROC 0.618), same 100 items. The reasoning-text approach separates the incorrect group's median above the correct group's; the digit-only approach ties them exactly. The visible gap is not statistically resolved, however: the paired difference is +0.083, 95% CI $[-0.064, +0.233]$ (§4.3).

cut (§9.1) – two different failure modes, same open status. A still-larger model, or a text-only model given the ground-truth transcription, also remain untested for either arm. (The baseline-uncertainty-method gap mentioned in earlier drafts of this limitation is resolved: entropy has a measured, resolved win over token confidence, §6.1.)

- **The perception capability gate is diagnosed on one model, not systematically characterized.** LLaVA-NeXT's MCQ-vs-free-response accuracy split (12.3% vs. 0.8%) is a clean, specific finding, but it rests on manual inspection of a handful of real samples plus one systematic recount, not a broader audit of failure modes across many models. It explains why this specific model cannot be used to test perception entropy; it is not a general theory of when a VLM will or will not clear the gate.

- **The single-dataset limitation is not resolved, and the attempt to resolve it found a structural obstacle.** The two viable public benchmarks of handwritten student errors both consist entirely of error-containing items (§10), so neither can support the balanced design this report's central pooled-versus-stratified argument depends on. On the one that was runnable, the model answered "error" even when stating it could not read the image, making the result a statement about response bias rather than about grading. Every quantitative claim here therefore still rests on FERMAT alone.

- **A pre-registered bar can be honoured exactly and still be the wrong bar.** §9.3's screen was registered before the run and reported as written – and one variant "passed" an absolute 0.70 threshold by 0.0023 with a CI of $[0.616, 0.781]$, while no variant's paired difference against baseline excluded zero. The threshold was inherited from confirmation runs with tight intervals and is too blunt for a screen at $n = 150$. The result is reported as a pass on the letter of the registration *and* as unresolved on the evidence that matters;

the honest conclusion is the latter. This is a limitation of the screen's design, not a post-hoc reinterpretation of its outcome – but the distinction is only visible because the paired test was run, which argues for registering screens on paired differences in future.

- **A high perception AUROC is not sufficient evidence of a genuine signal, even when every derived column recomputes correctly from raw data.** InternVL3's 0.915 (§9.1) is the clearest case in this report of a metric driven by an extreme bimodal failure mode (total generation incoherence vs. not) rather than graded uncertainty – and it would have been reported as this project's best perception result without the sensitivity cut applied to every number here. The mechanism itself (five retries landing on different incomplete derivation fragments) is characterized on one model only, from one full-population automated pass plus a handful of manually read examples, not independently verified against a larger sample the way the Phase 4 manual-inspection findings were.
- **The stratified reasoning hypothesis is now confirmed at both model sizes, not merely at 7B.** 3B's version was originally untestable (8 misgraded items among 150 error-containing ones, under the registered minimum); nothing in Phase 2 licensed a conclusion about 3B specifically. Both 7B's and 3B's versions of the same stratum, after drawing more error items, clear the power bar and confirm the hypothesis, landing close together (§7.4, §7.5).
- **The three unanimous (zero-entropy) misgraded items all have real errors invisible to arithmetic-checking, not label noise.** Two were initially misread as errorless from the pert_a text alone; checking the source images instead (§7.3) found an unsubstituted variable and a notation inconsistency, plus a third case (an unwarranted unit) in the 200-item extension. Read together, they suggest the model checks a solution's core computation and does not check whether its final answer statement is complete, consistent, or licensed by the problem – a narrower description of the ceiling than "cannot do arithmetic," but based on only three items and not independently verified against a larger sample.
- **Phase 1 numbers are exploratory.** With a minority class of 18–58 items, its CIs span roughly ± 0.12, and four of five comparisons were unresolvable. Several of its analyses were also chosen after seeing data. They are retained to document the path to the Phase 2 design, not as standing claims.
- **Confidence intervals were added retrospectively.** The experiments were designed, run, and initially written up without them; the intervals were computed afterwards on the same data. They correct the report's claims but cannot retroactively supply the statistical power the experiments were never sized for.
- **Perception correctness label is strict.** Transcription correctness requires the extracted final answer to match ground truth after canonicalization; some "incorrect" classifications are plausibly extraction-span mismatches (the model's answer and the ground truth landing in different canonicalization tiers) rather than genuine disagreement, which would mean the true accuracy is somewhat higher than 42%. This biases the reported accuracy conservatively, not favorably.
- **No human double-grading.** Ground-truth labels are taken as given from the dataset.
- **One post-hoc change was made to the Phase 2 analysis code.** The minority-class guard was applied to the inversion verdict after the run, correcting a branch that had been

skipping it. The registered threshold was not altered, but the change was made with the data visible and moved one verdict from "confirmed" to underpowered.

- **Reasoning entropy's ceiling is model- and task-specific, and three alternative prompt designs were screened and rejected.** Restating the result before judging, an explicit balanced-base-rate framing, and eliciting a 0–100 confidence were all tested at $n = 100$ (§6.3); none cleared the pre-registered screening bar. This does not rule out every possible prompt design, but it does rule out the simplest fixes, and the *restate* variant's behavior (a large response-distribution shift with no accuracy gain) argues for a genuine capability ceiling over a promptable artifact.

- **Reasoning-text clustering has a confirmed mechanical defect at $K = 15$.** Tested and found to collapse to near-random (AUROC 0.520) due to transitive cascade over-merging (§4.4). The mechanism is verified by direct inspection of cluster membership; the magnitude of its effect on aggregate AUROC is not well determined at this n. The underlying cross-encoder is also a small, general-purpose NLI model rather than one tuned for mathematical reasoning text, which may independently limit how far this approach can go even after the cascade issue is fixed.

14 Recommended Next Steps

Phases 4 and 5 resolved the items that used to head this list: the reasoning arm has a confirmed, sizable stratified signal, and it holds across both model sizes and, now, three model families tested (with InternVL3's magnitude weaker and unconfirmed, §9.2). What remains is narrower, and is being pursued toward a paper submission (see the working roadmap for scope, priority order, and the current external deadline).

1. **Find a model that is both accurate enough and gradedly uncertain enough to confirm perception entropy outside Qwen.** LLaVA-NeXT failed on accuracy (§8.1); InternVL3 cleared the accuracy bar but failed the sensitivity cut (§9.1) – two distinct disqualifications, not one recurring problem. This is still the single largest open item: the reasoning result has cross-family evidence, perception does not.

2. **Design screening bars around the interval, not the point estimate.** §9.3 ran its three-idea screen exactly as pre-registered and still produced an awkward outcome: one variant "passed" a 0.70 bar by 0.0023 with a CI running $[0.616, 0.781]$, while the paired comparison against baseline did not resolve for any of the three. The bar was sound for the confirmation runs it came from and too blunt for a screen at $n = 150$. A better-specified screen would register on the paired difference (does the variant beat baseline on the same items, CI excluding zero) rather than on an absolute point-estimate threshold – the same reasoning that makes `bootstrap_auroc_difference_ci` the right test everywhere else in this report.

3. **Decide whether the reasoning arm's stratified result is the headline finding or a secondary one.** It is now positive, confirmed, and holds across two model sizes *and* two model families, which changes the shape of this report's contribution: not "perception works, reasoning doesn't," but "entropy works on both arms once a model's own response bias is accounted for." Whether that reframing belongs in the abstract's emphasis or stays a Phase 4/5 result is an editorial choice, not a methodological one.

4. **The label-noise question is resolved: there isn't any, in the three cases checked.** §7.3 initially raised it for two misgraded items with no findable error in the pert_a text; a check against the source images (`pilot/07_manual_case_inspection.ipynb`) found real errors in both, plus a third case in the 200-item extension. A natural follow-up, not yet done: check whether the "core computation only" characterization holds on a larger sample than three items, since it was formed from exactly the cases this check happened to cover.

5. **Perception is a deployable result, and the deferral mechanism is now written up alongside the AUROC** (§5.6). It has AUROC 0.835, survives every artifact and scoring-strictness cut, holds across handwriting quality and image quality, beats an independent baseline by a resolved margin, and improves further with K. The remaining open item on this side is narrower than a full write-up: about two-thirds of conformal calibration splits could not reach a 30% risk target at all, because $K = 5$ gives too few operating points. $K = 10$ (§6.1) is the direct, already-tested fix; drawing it for the full $n = 300$ sample rather than the smaller confidence-baseline subset would let the deferral analysis use it too.

6. **The pairwise NLI matrix is already saved** for the $n = 300$ run, so any clustering-variant experiment – including a fix for the cascade fragility of §4.4 – is now a free offline computation, and now has a stratified positive result to try to sharpen rather than a null result to abandon.